A planar graph without four- or nine-cycles that is not three-colourable

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Abstract

A question of Lu, Wang, Wang, Bu, Montassier and Raspaud, as reproduced by Kang, Jin and Wang [2], asks which integers $i \in \{5, 6, 7, 8, 9\}$ have the property that every planar graph without cycles of length four or i admits a proper three-colouring. We answer the explicitly posed $i = 9$ case negatively by giving an 82-vertex planar graph with 136 edges, no four- or nine-cycles, and no proper three-colouring. The witness also has no ten-cycle. It complements the earlier disproof of Steinberg's $i = 5$ conjecture by Cohen-Addad, Hebdige, Král', Li and Salgado. The proof is computer-assisted: a supplied rotation system establishes planarity, two cycle enumerators check the forbidden lengths, and exact constraint elimination and a DRAT unsatisfiability certificate separately establish non-colourability. The graph, rotation system, formula, certificate and standalone checkers are supplied. The written reductions and executable checkers are not formalized in a proof assistant.

The proofs, the search code, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

Status: computer-assisted; not expert-reviewed; AI-assisted.

1 Introduction

All graphs in this paper are finite and simple. A proper three-colouring is a map $f : V(G) \to \{0, 1, 2\}$ with $f(u) \neq f(v)$ for every edge uv. A cycle is simple; it need not be induced. Thus forbidding a cycle of length k, denoted C_k, forbids cycles with chords as well.

Steinberg conjectured in 1976 that every planar graph without C_4 or C_5 is properly three-colourable; see the account in [4]. Lu, Wang, Wang, Bu, Montassier and Raspaud [1] posed a broader forbidden-pair question, as reproduced in Kang, Jin and Wang's Problem 1.2 [2]: for which $i \in \{5, 6, 7, 8, 9\}$ is every $\{C_4, C_i\}$-free planar graph properly three-colourable? The $i = 5$ case is Steinberg's conjecture. Cohen-Addad, Hebdige, Král', Li and Salgado disproved that case in a preprint posted in April 2016, published in 2017 [4]. Jin, Kang, Liu and Wang's 2020 preprint [3] explicitly describes the cases $i \in \{6, 7, 8, 9\}$ as open. The result here answers $i = 9$.

Theorem 1. *There exists an 82-vertex planar graph with 136 edges, without cycles of length four or nine, that is not properly three-colourable.*

This is a negative answer to the explicitly posed $i = 9$ case. The particular graph also excludes C_{10}, an additional checked property recorded in the cycle table below. It contains C_5, C_6, C_7 and C_8, so it gives no information on $\{C_4, C_5, C_9\}$- or $\{C_4, C_8, C_9\}$-free planar graphs, nor on the cases $i \in \{6, 7, 8\}$. No assertion about minimum order is made.

The earlier construction does not itself give this exclusion: its G_1 component in Figure 1 [4] contains the nine-cycle $a, f, k, n, m, j, i, h, g, a$ in the paper's own labels. This component is retained in the published construction.

The distinction from neighbouring positive theorems matters. Kang, Jin and Wang prove proper three-colourability when C_6 is also excluded [2]. The $\{C_4, C_7, C_9\}$-free class is properly three-colourable by Lu and coauthors, as also recorded in [2]; Kang, Jin and Zhu prove DP-3-colourability of that class [5]. DP-3-colourability implies three-choosability, which implies ordinary proper three-colourability. Consequently, failure of a list or DP colouring property alone would not supply the obstruction required here. Defective colourings are a different notion: a (d_1, d_2, d_3)-colouring permits each colour class to induce maximum degree at most d_j. The $(1, 1, 0)$ result of Dai, Wang and Xu [6] and the $(2, 0, 0)$ result of Chen, Dai, Nie, Wang and Yu [7] concern the exact forbidden pair but allow monochromatic edges. They are compatible with Theorem 1. Xiao and Huang's weak DP-$(\mathcal{F}_2, \mathcal{F}_2, \mathcal{F}_0)$-colouring theorem for this pair uses classes $\mathcal{F}_d$ of forests of maximum degree at most d [8]; it likewise does not assert proper three-colourability.

The accessible question statement used here is the reproduction in [2]. The defective and weak DP statements cited above are read through publisher abstracts or excerpts rather than their full papers. Literature coverage has further gaps: the full texts of Ang–Leo's *Exploring the Weak Steinberg's Conjecture*, the updated Bu–Wang–Wang survey *On Coloring Problems*, and Li–Niu–Tian's *On the strong Bordeaux Conjecture* were unavailable; their contents are not ruled out by this account [9, 10, 11]. The accessible abstract of the first concerns $\{4, 5, 6\}$. These limits could affect priority or attribution; they do not affect the fixed witness or its checks.

The supplied bundle contains the graph and rotation system reproduced in Appendix A, with the formula, certificate and checkers at the relative paths specified in Sections 4–5. The two embedding and cycle implementations are separate Python and C programs. The two non-colourability paths are exact constraint elimination over the three colour values and a DRAT certificate checked by DRAT-trim. The mathematical reductions and ordinary executable programs, including the CNF generator, remain part of the proof's trust base.

2 The graph and its embedding

The input `data/graph.txt`, reproduced in Appendix A, defines G on vertices $0, \ldots, 81$ by 136 unordered edges. The accompanying `data/rotation.txt` gives a cyclic order of the neighbours at each vertex. The two planarity implementations separately validate the edge count, simplicity, neighbour lists and connectivity.

Let $D = \{(u, v) : uv \in E(G)\}$ be the darts. Write $\rho_v(u)$ for the neighbour following u in the cyclic order at v. The face permutation is

$$\phi(u, v) = (v, \rho_v(u)).$$

Both implementations count 56 orbits of ϕ. To justify this test, place an oriented disc at every vertex, attach an untwisted band for each edge according to the cyclic orders, and cap each boundary walk with a disc. The result is a connected closed orientable surface with a cellular embedding of G. The boundary walks are precisely the orbits of ϕ, so its Euler characteristic is

$$|V| - |E| + |F| = 82 - 136 + 56 = 2.$$

By the classification of closed orientable surfaces this surface is the sphere. Deleting a point outside the embedded graph gives a plane embedding. This proves planarity; no geometric drawing or numerical coordinate tolerance is involved.

3 Excluding cycles

The Python enumerator grows every simple path from its smallest vertex s, using only other vertices greater than s. It records a closure to s when the path has at least three vertices and its second vertex is smaller than its last. Every undirected simple cycle has a unique smallest vertex and exactly two orientations, exactly one of which meets this last test. Thus each cycle is counted once.

The C implementation instead chooses an edge (s, t) with $s < t$, and enumerates simple s–t paths avoiding that edge, with all other vertices greater than s. Closing by ts gives a simple cycle. A cycle whose smallest vertex is s is found twice, once for each of its two edges at s; division by two gives the count. The implementation checks that each raw count is even. Both implementations run through length 14 and return the same table:

length	3	4	5	6	7	8	9	10	11	12	13	14
count	27	0	22	55	44	11	0	0	64	320	832	1824

In particular there is no C_4, C_9 or C_{10}. Since the maximum degree of this input is 10, even the coarse bound $82 \cdot 10^{14} < 2^{63}$ on raw enumeration counts is below the signed 64-bit limit used in C. All Python counts use arbitrary precision integers.

4 Two checks of non-colourability

4.1 Exact constraint elimination

For each edge uv start with the relation $R_{uv} = \{(a, b) \in \{0, 1, 2\}^2 : a \neq b\}$. Maintain a conjunction of relations on the remaining vertices. To eliminate v, collect all relations mentioning it, let S be the union of their scopes minus v, and replace them by the relation

$$R_S = \{a \in \{0, 1, 2\}^S : \exists c \in \{0, 1, 2\} \bigwedge_{R \text{ mentioning } v} R(a, v = c)\}.$$

All other relations remain. The new conjunction is exactly the existential projection of the old one. Induction on eliminations shows that satisfiability is preserved. An empty relation proves that no colouring exists; a nonempty relation on the empty scope is a tautology and may be discarded.

The implementation enumerates all $3^{|S|}$ assignments and all three values of c. It chooses a vertex of minimum degree in the current interaction graph, breaking ties by vertex label. This choice affects cost, not correctness. On the specified graph it reaches an empty relation; the largest remaining scope S encountered has size three. This route imports no SAT solver, CNF generator or DRAT checker.

4.2 Exact graph-to-CNF reduction and symmetry

Independently introduce Boolean variables $x_{v,c}$, numbered $3v+c+1$, for $0 \leq v < 82$ and $c \in \{0, 1, 2\}$. The generator adds the clauses

$x_{v,0} \vee x_{v,1} \vee x_{v,2}$	for each v,
$\neg x_{v,a} \vee \neg x_{v,b}$	for each v and $0 \leq a < b < 3$,
$\neg x_{u,c} \vee \neg x_{v,c}$	for each $uv \in E(G)$ and c.

The first two lines specify exactly one colour at each vertex, and the third specifies distinct colours across every edge. Assigning the unique true colour is therefore a bijection between satisfying assignments of these clauses and proper three-colourings of G.

The lexicographically first triangle is $(0, 1, 6)$. The generator also adds $x_{0,0}$, $x_{1,1}$ and $x_{6,2}$, that is, unit clauses 1, 5 and 21. Every proper colouring gives distinct colours to this triangle. There is a permutation of the three colour names taking them, in this order, to $0, 1, 2$; the other clauses are invariant under this permutation. Adding these units thus preserves satisfiability. No graph automorphism or unproved assertion about the search is used.

The resulting formula has $3 \cdot 82 = 246$ variables and $4 \cdot 82 + 3 \cdot 136 + 3 = 739$ clauses. The supplied `data/col3.cnf` is compared with the generated formula by parsed header and ordered clauses, ignoring comments. The certificate `data/col3.drat`, generated with CaDiCaL [13], is checked by DRAT-trim [12]. It contains 97 addition/deletion steps in standard text DRAT, ending with the empty clause. Its ordered steps are a lossless transcription of the binary proof. DRAT's checked redundancy conditions preserve satisfiability under additions; deletions only weaken the formula. Derivation of the empty clause therefore proves unsatisfiability of the input. This permits replay without trusting a solver's bare unsatisfiability answer or depending on a solver version. Together with the symmetry argument and the exact reduction above, this separately proves that G has no proper three-colouring.

5 Replay, controls and scope

The complete standalone check is `sh checks/check.sh`, run from the bundle root. It needs Python 3.10 or later, a C compiler and a POSIX shell; there are no third-party Python packages or network downloads. It compiles the supplied C sources in a temporary directory, runs both embedding checks, both cycle enumerators, elimination, CNF comparison and DRAT verification. The DRAT route requires successful termination and the exact success status, not just an absent error message.

The same command exercises deliberate corruptions. Deleting edge $(0, 1)$ must make the elimination result colourable and invalidate the original DRAT proof on the modified formula. Adding $(0, 12)$ creates the cycle $0, 6, 60, 12, 0$; adding $(0, 7)$ creates $0, 1, 6, 60, 12, 62, 63, 13, 7, 0$. Both cycle implementations must reject each modified graph. Swapping two neighbours in the rotation at zero changes the Euler characteristic to zero and must be rejected by both embedding checks. This is rejection of a rotation, not a non-planarity claim for the underlying graph. Finally a proper prefix containing only the first 45 complete proof steps must fail DRAT verification. These are nine control outcomes. The certificate checks and the reductions in Sections 2–4 establish Theorem 1.

This is a computer-assisted proof, not a proof-assistant formalization. The rotation-system argument, enumeration completeness, CNF reduction, colour symmetry and constraint-elimination argument are mathematical arguments written here. Their implementations and DRAT-trim run as ordinary programs and are not themselves formally verified. The different implementations and non-colourability methods reduce shared implementation risk. The witness was found by composing small planar graphs with designated contact vertices, tabulating the possible colour patterns on those contacts, and seeking a composition with no allowed pattern. This explains the discovery route; the certificate checks apply to the final edge list directly. The search is not needed for the proof and makes no claim of exhaustive search, optimal order or applicability to another forbidden pair.

Credits and licence. Credit for posing and developing the forbidden-cycle questions belongs to the researchers cited above, and for the earlier Steinberg disproof to Cohen-Addad and coauthors. The SAT certificate uses CaDiCaL and is checked with Heule and Wetzler's DRAT-trim. This work, including the paper, is under Apache-2.0 (the repository's root `LICENSE`); the unmodified DRAT-trim source retains MIT. The full applicable notices and third-party provenance appear in

`THIRD_PARTY.md`.

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A Explicit graph and rotation

The graph list starts with the order and size, followed by every edge. The rotation gives every neighbour in cyclic order. These are the same input files used by the standalone check; no reconstruction from a drawing is necessary.

Graph

```
82 136
0 1
0 2
0 3
0 4
0 5
0 6
1 6
1 18
1 19
1 20
1 21
2 23
2 32
3 24
3 33
4 28
4 34
5 29
5 35
6 60
6 61
7 8
7 9
7 10
7 11
7 12
7 13
7 14
7 15
7 16
7 17
8 36
8 40
9 37
9 41
10 42
10 44
11 43
11 45
12 60
12 62
13 61
13 63
14 68
14 72
15 69
15 73
16 74
16 76
17 75
17 77
18 51
18 64
19 52
19 65
20 56
20 66
21 57
21 67
22 23
22 24
22 25
22 26
23 32
24 33
25 26
25 38
25 39
26 42
26 43
27 28
27 29
27 30
27 31
28 34
29 35
30 31
30 36
30 37
31 46
31 47
32 33
34 35
36 40
37 41
38 46
38 48
39 47
39 49
40 41
42 44
43 45
44 45
46 48
47 49
48 49
50 51
50 52
50 53
50 54
51 64
52 65
53 54
53 70
53 71
54 74
54 75
55 56
55 57
55 58
55 59
56 66
57 67
58 59
58 68
58 69
59 78
59 79
60 62
61 63
62 63
64 65
66 67
68 72
69 73
70 78
70 80
71 79
71 81
72 73
74 76
75 77
76 77
78 80
79 81
80 81
```

Rotation

```
0: 1 3 2 5 4 6
1: 0 6 19 18 21 20
2: 23 0 32
3: 33 0 24
4: 28 0 34
5: 35 0 29
6: 1 0 60 61
7: 12 8 9 11 10 14 15 17 16 13
8: 7 36 40
9: 37 7 41
10: 42 7 44
11: 45 7 43
12: 60 7 62
13: 7 61 63
14: 7 68 72
15: 69 7 73
16: 74 7 76
17: 77 7 75
18: 51 1 64
19: 65 1 52
20: 56 1 66
21: 67 1 57
22: 25 23 24 26
23: 22 2 32
24: 3 22 33
25: 38 22 26 39
26: 22 42 43 25
27: 30 28 29 31
28: 27 4 34
29: 5 27 35
30: 36 27 31 37
31: 27 46 47 30
32: 2 33 23
33: 32 3 24
34: 4 35 28
35: 34 5 29
36: 8 30 40
37: 30 9 41
38: 46 25 48
39: 25 47 49
40: 41 8 36
41: 9 40 37
42: 26 10 44
43: 11 26 45
44: 10 45 42
45: 44 11 43
46: 31 38 48
47: 39 31 49
48: 49 46 38
49: 47 48 39
50: 53 51 52 54
51: 50 18 64
52: 19 50 65
53: 70 50 54 71
54: 50 74 75 53
55: 58 56 57 59
56: 55 20 66
57: 21 55 67
58: 68 55 59 69
59: 55 78 79 58
60: 6 12 62
61: 13 6 63
62: 63 60 12
63: 61 62 13
64: 18 65 51
65: 64 19 52
66: 20 67 56
67: 66 21 57
68: 14 58 72
69: 58 15 73
70: 78 53 80
71: 53 79 81
72: 73 14 68
73: 15 72 69
74: 54 16 76
75: 17 54 77
76: 16 77 74
77: 76 17 75
78: 59 70 80
79: 71 59 81
80: 81 78 70
81: 79 80 71
```